

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Winter Examination – 2022****Course: B. Tech. Branch: Information Technology Semester: VII****Subject Code & Name: BTITE703B Soft Computing****Max Marks: 60****Date: 01-02-2023****Duration: 3 Hr.****Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Draw and compare Biological and Artificial neuron structure.	L-2/CO-1	6
B) Implement AND function using McCulloch-Pitts neuron model. (Use binary data representation).	L-3/CO-1	6
C) List any four activation functions with their equations and graphs.	L-1/CO-1	6
Q.2 Solve Any Two of the following.		12
A) Explain the architecture of Multilayer perceptron?	L-2/CO-2	6
B) What is backpropagation? How does backpropagation work?	L-3/CO-2	6
C) Describe in brief RBFN-Radial basis function network?	L-2/CO-2	6
Q. 3 Solve Any Two of the following.		12
A) List and explain fuzzy membership functions?	L-2/CO-3	6
B) Find following fuzzy operations on given fuzzy sets.	L-3/CO-3	6
$\underline{A} = \{(x_1, 0.2), (x_2, 0.5), (x_3, 0.6), (x_4, 0.8), (x_5, 1.0)\}$ $\underline{B} = \{(x_1, 0.8), (x_2, 0.6), (x_3, 0.4), (x_4, 0.2), (x_5, 0.1)\}$		
i) $A \cup B$ ii) $A \cap B$ iii) A^C iv) $A^C \cup B^C$		
C) Explain architecture of fuzzy logic with suitable diagram	L-2/CO-3	6
Q.4 Solve Any Two of the following.		12
A) Categorize the different methods of defuzzification process?	L-4/CO-4	6
B) Explain assumptions in a Fuzzy Control System Design.	L-2/CO-4	6

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| C) | List the five functional blocks FIS- Fuzzy Inference System with a block diagram and working. | L-2/CO-4 | 6 |
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Q. 5	Solve Any Two of the following.		12
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| A) | Design membership function for Dirt, Grease and Wash time . For a Fuzzy logic controller for Domestic washing machine to determine Wash time using two inputs Dirt and Grease on cloths. Use 3 descriptor for input variables and five descriptors for output variable as follows,
Dirt= {small dirt, medium dirt, large dirt}
Grease= {No grease, medium grease, large grease}
Wash Time= {Very short, short, Medium, Large, Very Large} | L-6/CO-5 | 6 |
| B) | Write advantages and limitations of ANFIS- adaptive neuro fuzzy inference systems. | L-2/CO-5 | 6 |
| C) | Draw and explain architecture of ANFIS- adaptive neuro fuzzy inference systems. | L-2/CO-5 | 6 |

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